Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/16/2006
PI Name:	Weber, Richard Ph.D.		
Project Title:	Microgravity Studies of Liquid-Liquid Phase Transitions in Undercooled Alumina-Yttria Melts		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Glasses and ceramics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	05/01/2003	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: project added 9/2006 when first became aware of it		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nordine, Paul (Containerless Research Inc.)		
Grant/Contract No.:	NNM04AA23G		
Performance Goal No.:			
Performance Goal Text:			

The scientific objective of this research is to increase the fundamental knowledge base for liquid-phase processing of technologically important oxide materials. The experimental objective is to define conditions and hardware requirements for microgravity flight experiments to test and expand the experimental hypotheses that:
1. Liquid phase transitions can occur in undercooled melts by a diffusionless process.

2. Onset of the liquid phase transition is accompanied by a large change in the temperature dependence of melt viscosity.

Experiments on undercooled YAG (Y3Al5O12)- and rare earth oxide aluminate composition liquids demonstrated a large departure from an Arrhenian temperature dependence of viscosity. Liquid YAG is nearly inviscid at its 2240 K melting point. Glass fibers were pulled from melts undercooled by ca. 600 K indicating that the viscosity is on the order

Task Description:	of 100 Pa s (1000 Poise) at 1600 K. This value of viscosity is 500 times greater than that obtained by extrapolation of data for temperatures above the melting point of YAG. These results show that the liquids are extremely fragile and that the onset of the highly non-Arrhenian viscosity-temperature relationship occurs at a temperature considerably below the equilibrium melting point of the solid phases. Further results on undercooled alumina-yttria melts containing 23-42 mole % yttrium oxide indicate that a congruent liquid-liquid phase transition occurs in the undercooled liquids. The rates of transition are inconsistent with a diffusion-limited process. This research is directed to investigation of the scientifically interesting phenomena of polyamorphism and fragility in undercooled oxide liquids. The results bear on the technologically important problem of producing high value rare earth-based optical materials. In addition to the work directly related to alumina-yttria binaries, there was considerable work performed on alumina-calcia and magnesia-silica binary materials that are of interest in connection with fragile liquids and polyamorphism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developed advanced IR transmitting glass materials that are being commercialized through separate work. See, for example, NASA Tech Briefs, 30, 18a (January 2006).
Task Progress:	The research was mainly focused on the study of undercooled pseudo-binary alumina-yttria, alumina-calcia and magnesia-silica melts. Containerless techniques were being used to access non-equilibrium liquids and allow precise control of melt chemistry at high temperatures. Although the oxides of interest are extremely stable and exhibit only small oxygen composition changes, some properties of melts are very sensitive to the ambient oxygen pressure which must be controlled in order to investigate subtle effects in liquid oxides. Properties under investigation were the onset and kinetics of phase transitions in the liquid and the structure and viscosity of the undercooled liquid as a function of the temperature. Containerless experiments using aero-acoustic (AAL), aerodynamic (CNL) and electrostatic (ESL) levitation techniques were employed to achieve precise control of melt chemistry, access highly undercooled melts and glassy states, and control the effects of interaction with container walls on the evolution of morphology. The containerless experiments employ Containerless Research Inc.'s aero acoustic and aerodynamic levitation facilities with CO2 laser beam heating to investigate the effects of processing conditions on the compositions of interest. In these experiments, the melts are highly stirred by the aerodynamic and acoustic levitation forces and equilibrate rapidly with the process atmosphere. The experiments were performed in atmospheres with a controlled p(O2), which: (i) preserves the melt stoichiometry and avoids partial reduction of the oxides, and (ii) achieves conditions under which the liquid phase separation and glass formation occur. Activities and Results: The major activities were to: 1. Investigation of liquid oxides properties, structure and processing behavior. 2. Publish/present results. More detailed reports are provided in previous NASA reports, the NASA Taskbook, and our published papers - see bibliography.
Bibliography Type:	Description: (Last Updated: 06/06/2026)
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